Nova Analytics - EDA Report

Dataset: insurance (1).csv

1,338 rows x 7 columns

Dataset Information

Shape

Rows	**1,338**
Columns	**7**

Data Types

Column	Data Type	Non-Null	Null	Unique
age	int64	1,338	0	47
sex	object	1,338	0	2
bmi	float64	1,338	0	275
children	int64	1,338	0	6
smoker	object	1,338	0	2
region	object	1,338	0	4
expenses	float64	1,338	0	1337

Descriptive Statistics

Column	Mean	Median	Std Dev	Min	25%	50%	75%	Max
age	39.21	39.00	14.05	18.00	27.0	39.0	51.0	64.00
bmi	30.67	30.40	6.10	16.00	26.3	30.4	34.7	53.10
children	1.09	1.00	1.21	0.00	0.0	1.0	2.0	5.00
expenses	13270.42	9382.03	12110.01	1121.87	4740.29	9382.03	16639.92	63770.43

Missing Value Analysis

Total Missing Values: **0**

Column	Missing Count	Percentage
age	0	0.0%
sex	0	0.0%
bmi	0	0.0%
children	0	0.0%
smoker	0	0.0%
region	0	0.0%
expenses	0	0.0%

No missing values detected in this dataset.

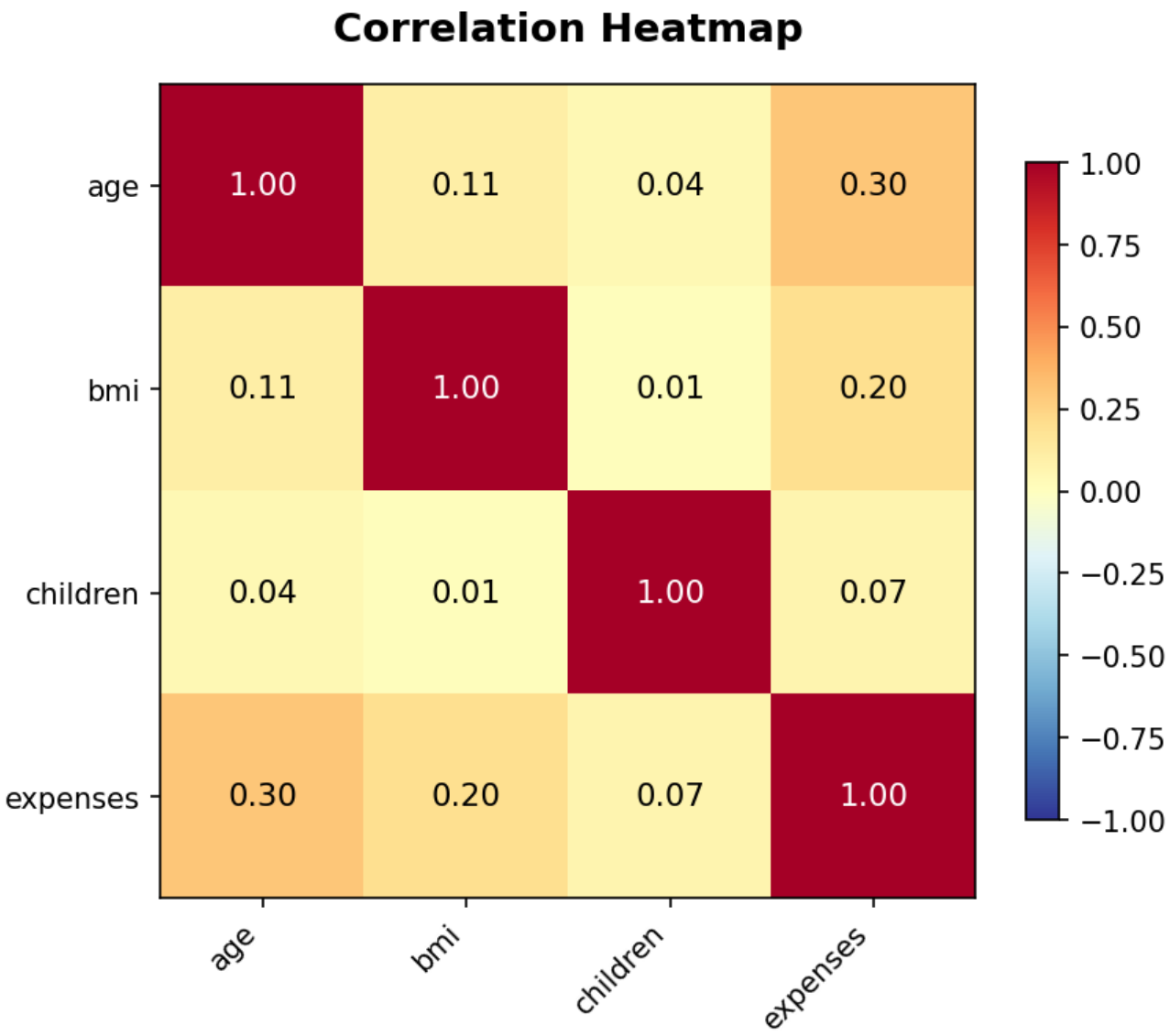
Duplicate Analysis

Duplicate Rows	**1 (0.1%)**
Total Rows	**1,338**

Correlation Analysis

Correlation Matrix

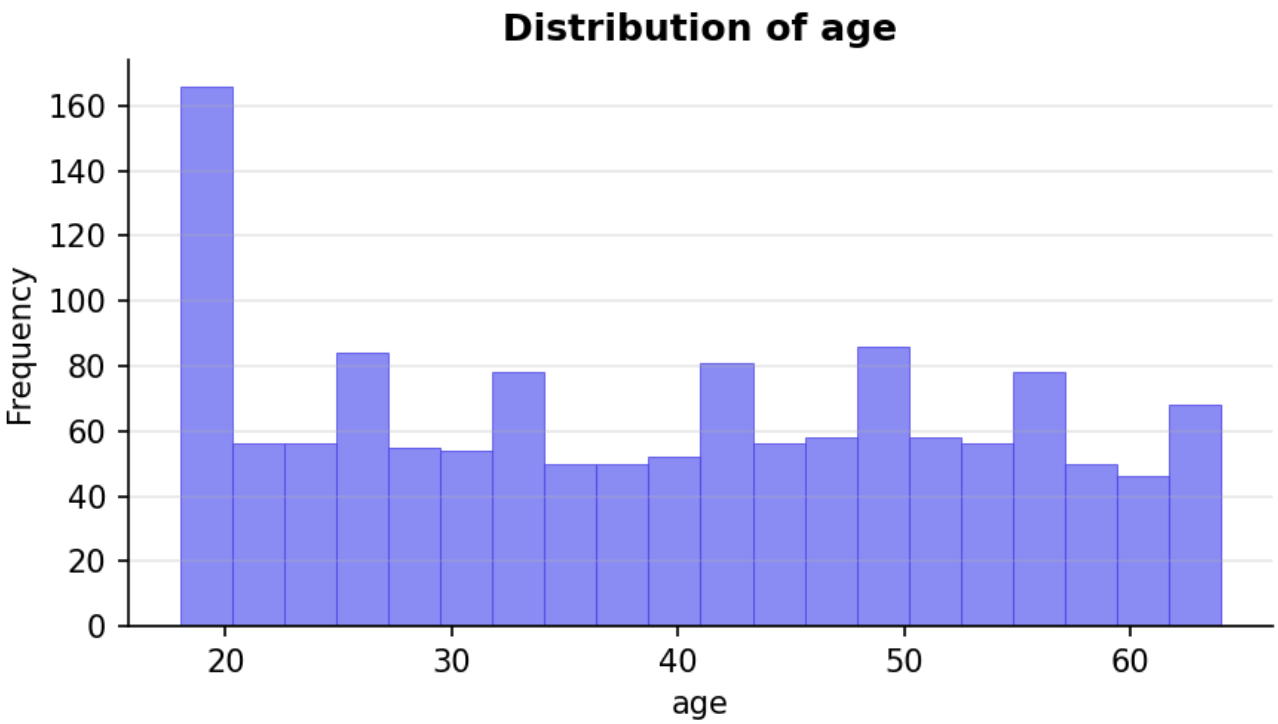
	age	bmi	children	expenses
age	1.00	0.11	0.04	0.30
bmi	0.11	1.00	0.01	0.20
children	0.04	0.01	1.00	0.07
expenses	0.30	0.20	0.07	1.00

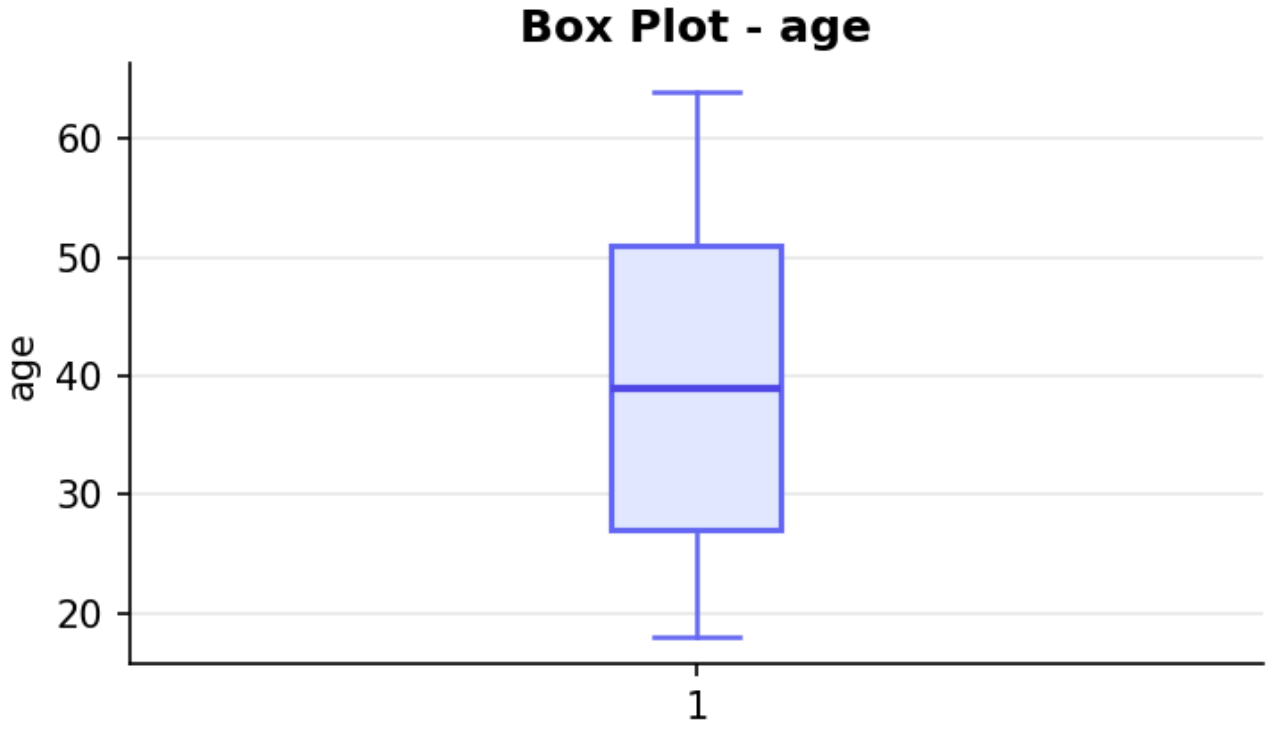


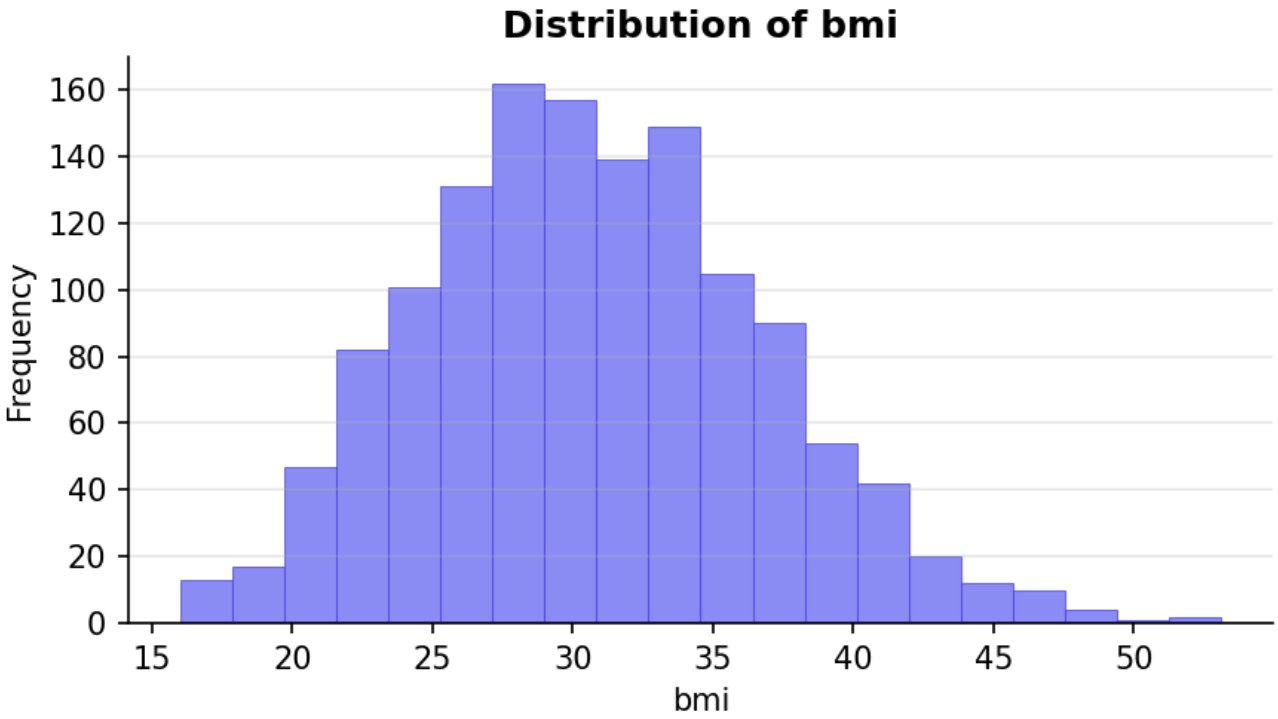
Outlier Analysis

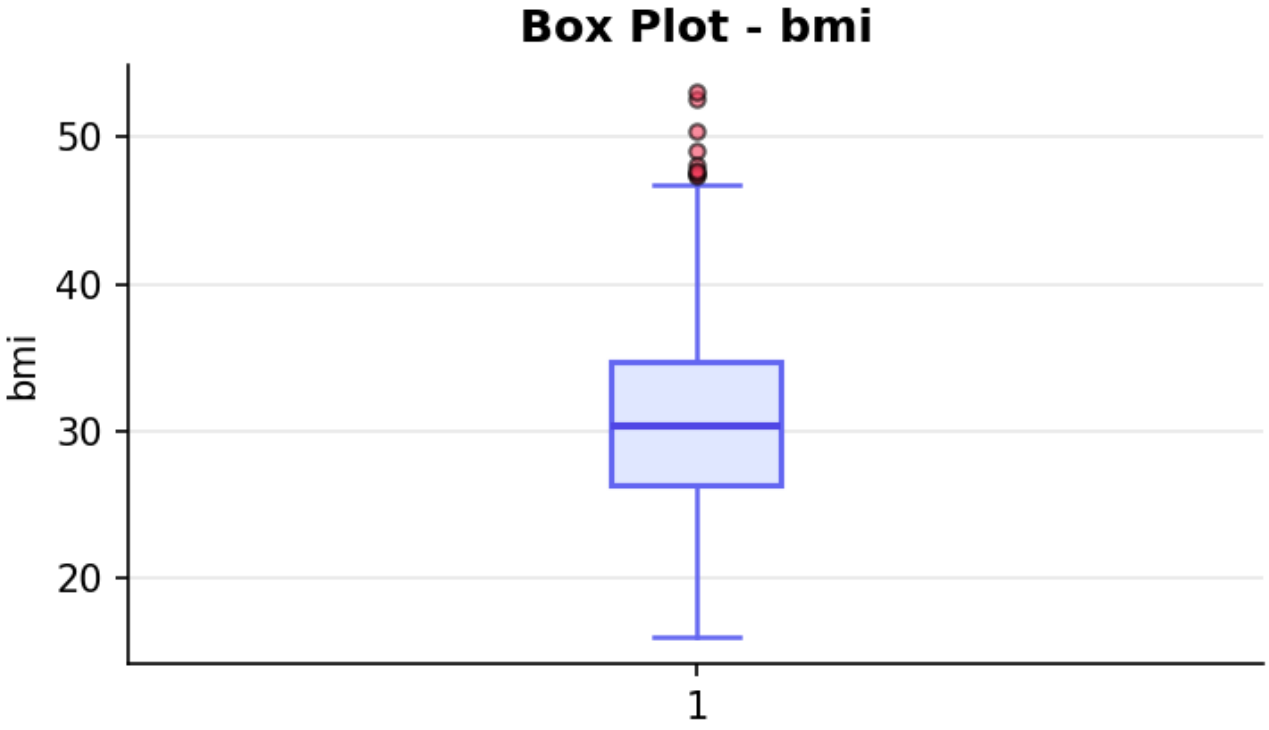
Column	Outliers	%	Q1	Q3	IQR	Lower Bound	Upper Bound
age	0	0.0%	27.00	51.00	24.00	-9.00	87.00
bmi	9	0.7%	26.30	34.70	8.40	13.70	47.30
children	0	0.0%	0.00	2.00	2.00	-3.00	5.00
expenses	139	10.4%	4740.29	16639.92	11899.63	-13109.15	34489.36

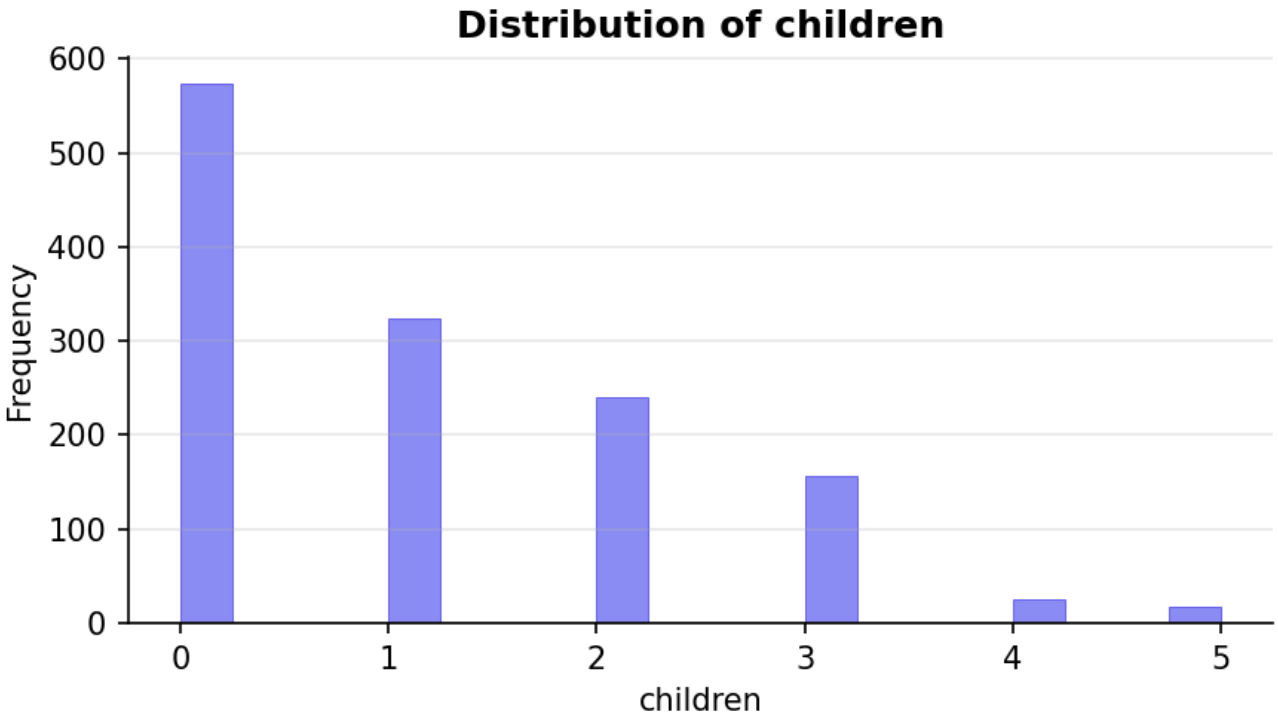
Distribution Analysis

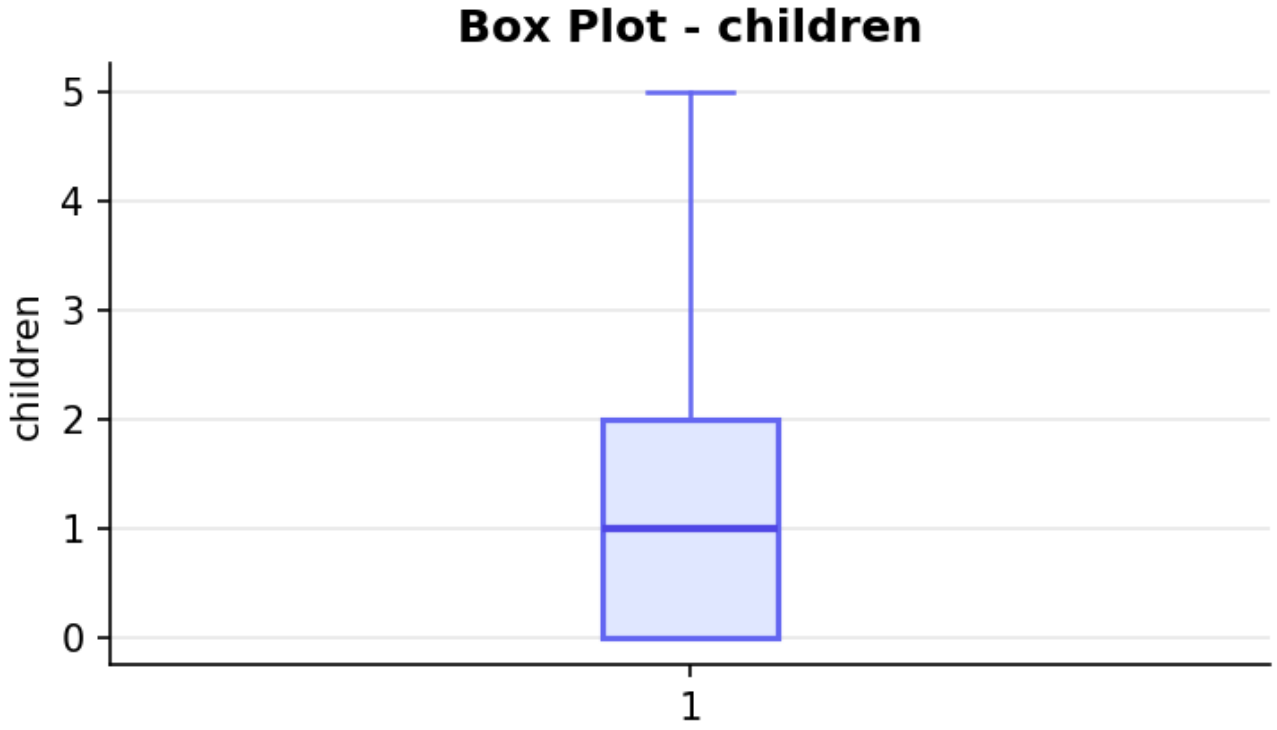

Distribution of age
Frequency
160
140
120
100
80
60
40
20
0
20
30
40
50
60
age

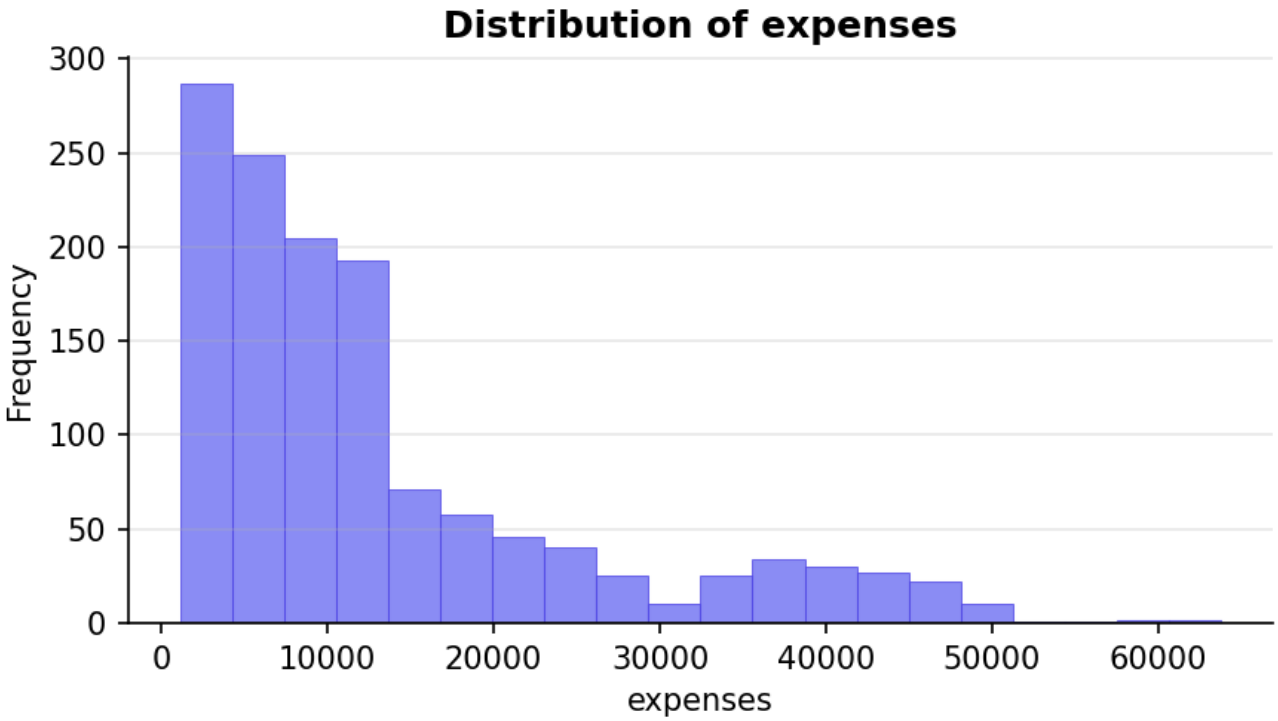

Box Plot - age
age
60
50
40
30
20
1

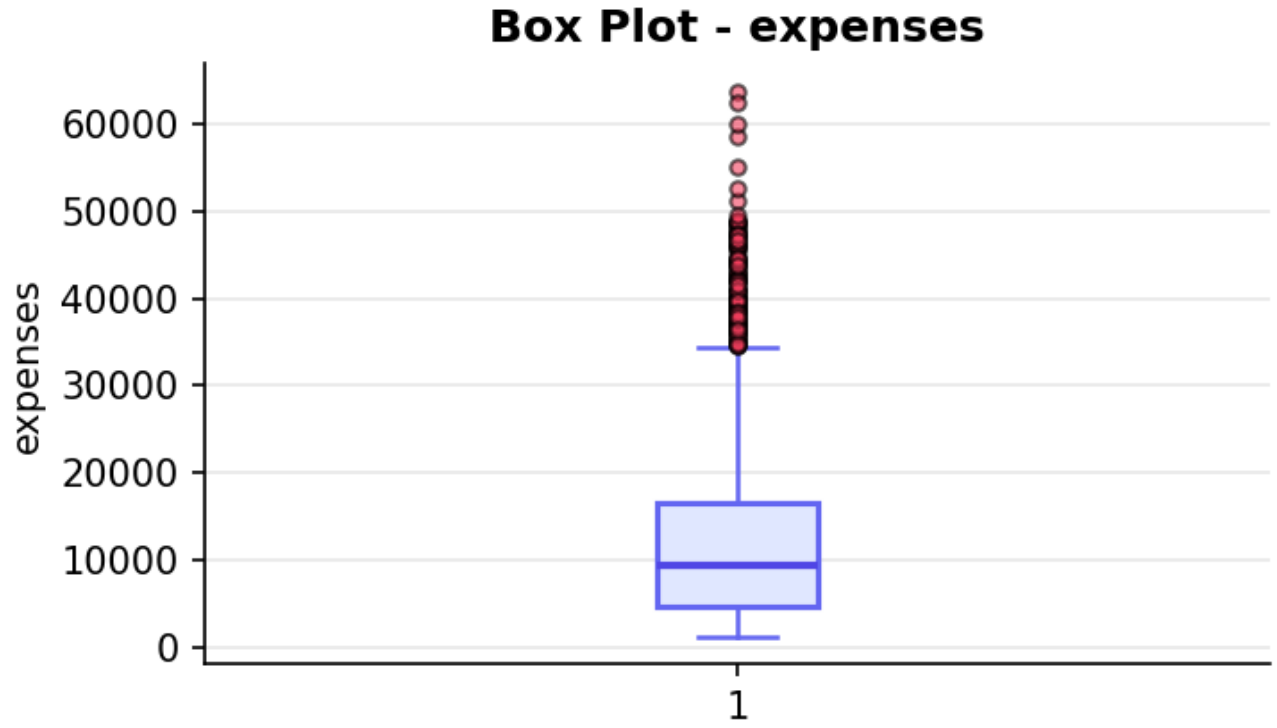

Distribution of bmi
Frequency
160
140
120
100
80
60
40
20
0
15
20
25
30
35
40
45
50
bmi


Box Plot - bmi
bmi
50
40
30
20
1


Distribution of children
600
500
400
300
200
100
0
Frequency
0
1
2
3
4
5
children


Box Plot - children
5
4
3
2
1
0
children
1


Distribution of expenses
Frequency
300
250
200
150
100
50
0
0
10000
20000
30000
40000
50000
60000
expenses


Box Plot - expenses
expenses
60000
50000
40000
30000
20000
10000
0
1